# Exit Ticket · reflect and retrieve

## Exit Ticket

Getting a correct output once guarantees an algorithm is bug-free.

## Use key vocabulary from memory

1. Explain mark scheme in your own words.

2. Explain termination in your own words.

3. Explain auxiliary space in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 309 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Paper 2 examiner-style feedback

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Why must recursive calls progress?

### Q2

What does a midpoint calculate?

### Q3

What happens when low>high?

### Q4

C# [2,4,6] with a[1]=a[0]+a[2]. Final array?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Understand — Identify whether the error lies in problem reading, algorithm state or pseudocode.

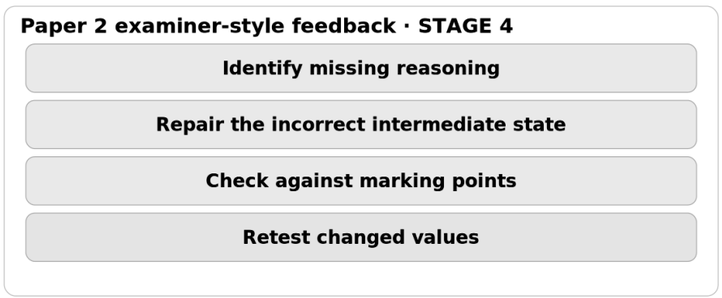
## Core idea 2

Apply — Explain every bound update, recursive return or data-structure operation.

## Core idea 3

Evaluate — Retest the method with a different input to demonstrate transfer.

**Key visual · complete concept**








# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Correct: 'merge sort has O(1) auxiliary space because the temp array is short-lived'. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

### Question 1

An algorithm's recursion has no base case. Explain fault and give two repairs. [6]

### Question 2

Trace binary search for 9 in [1,3,5,7,9], recording every low, high and midpoint. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

### Vocabulary · use the correct technical terms

mark scheme — Criteria for awarding marks.

termination — Algorithm eventually stops.

auxiliary space — Additional storage beyond the input.

### Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Repair a SumTo recursion missing a base case.

## Follow the worked method

1. Explain the result — The n==0 base case stops recursion and returns zero.

**Source model · final illustrated step**

**IDO · METHOD**

**The n==0 base case stops recursion and returns zero.**

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Explain the result
The boundaries exclude a failed midpoint, so the remaining interval shrinks.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.